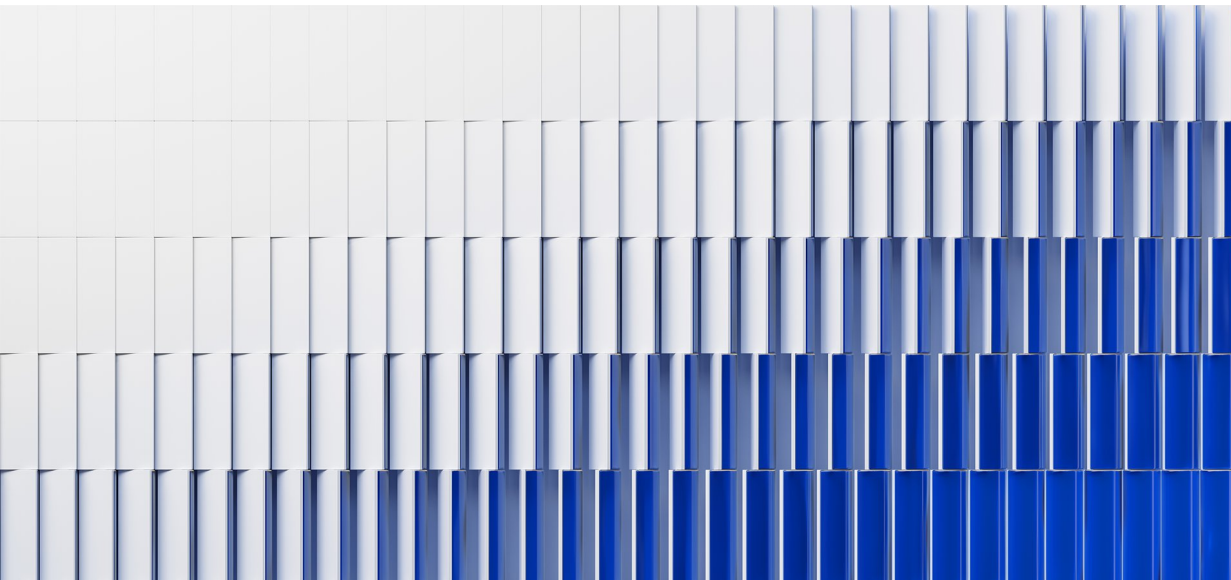


IBM Storage Ceph versus Community Ceph



Andrés Arroyo Samayoa

Cloud Management/SW Solution Architect

aarroyo@gbm.net



Open-source Ceph Storage

Open-source Ceph Storage has become [a leading choice](#) for organizations seeking scalable, software-defined storage.

While the appeal of a free, self-managed solution is strong during early adoption, the evolving needs of mature infrastructure can reveal hidden risks and operational burdens.

These challenges are often underestimated until they impact business reliability and growth.

'Free' is not costless
DIY storage brings ongoing complexity.



Key Comparison Areas

1. Support and Risk Ownership
2. Security and Compliance
3. Operational Overhead
4. Strategic Alignment

Ceph support

When it comes to support, Community Ceph does not provide formal assistance or service-level guarantees.

If problems arise, your IT team must troubleshoot independently, relying on community forums, IRC channels, or mailing lists.

This results in unpredictable response times, and resolutions can be delayed for weeks or even months, impacting productivity.

IBM Storage Ceph delivers 24/7 support, proactive monitoring, and expert access.

1

Direct line to
enterprise technical
experts

2

Automated issue
detection and
prevention

3

Proactive
monitoring with Call-
Home capabilities

4

Rapid escalation
and resolution of
critical incidents

Proactive risk mitigation

The distinction goes beyond faster problem resolution.

With IBM Storage Ceph, you reduce the risk of outages, data loss, and extended downtime.

These risks can have direct consequences on business operations, disrupt service delivery, and damage your organization's reputation in the marketplace.

Security Management

Security management is a significant differentiator.

Community Ceph places the onus on users for all patching, vulnerability assessments, and dependency checks.

Without automated or managed security controls, there is no assurance that your storage environment is protected against evolving threats or compliance issues.

Enterprise risk management frameworks and operational best practices.

Continuous security scanning by IBM and Red Hat teams

Secure delivery of all software and container images

Rapid updates for newly discovered vulnerabilities

Enterprise-grade compliance with industry standards



Certified compatibility

Modern enterprises need seamless integration and certified compatibility across hybrid and multi-cloud environments.

IBM Storage Ceph features [formal certifications](#) and direct integration support, making it convenient to ensure reliable interoperability with other mission-critical enterprise systems and technologies.



IBM Storage Ceph: Certified, Supported, and Compliant Integrations

1

Validated for
VMware,
Microsoft, and Red
Hat platforms

2

Accelerates
troubleshooting with
certified
compatibility

3

Ensures compliance
with industry
regulations and
standards

4

Maintains full
eligibility for vendor
support across your
stack

Feature maturity

Feature maturity is crucial when considering Ceph for production workloads.

The open-source Ceph community project contains a wide array of features, but not all are equally stable or fully maintained.

Some may still be classified as technology previews, while others may be deprecated, introducing potential risk if deployed in critical environments.

Quality Engineering, Documentation and Real-world testing

IBM Storage Ceph includes only those features that have passed stringent quality engineering, documentation, and real-world testing.

Immature or experimental features are clearly labeled and restricted from production use, shielding your workloads from instability and unexpected issues that could compromise reliability or performance.

Upstream Ceph
Code by IBM
Engineers

80%+

Customer Influence
on Roadmap

High

Client Responsibility

Deploying Community Ceph means you are responsible for hardware selection, initial configuration, and ongoing lifecycle management.

Errors in these areas can result in performance bottlenecks, system outages, or expensive rework, adding significant risk and overhead to your storage deployment.

IBM Storage Ceph Deployment options

Ready Nodes	Pre-validated Hardware	Accelerated Deployment
Ceph-as-a-Service	Managed Operations	Predictable Performance
Optimized Packages	Eliminate Guesswork	Consistent Results
Day-One Support	Expert Guidance	Reduced Risk

Community Ceph environments require continuous attention

Ongoing operations with Community Ceph can be demanding.

Your team manages upgrades, scaling, monitoring, disaster recovery, and incident response.

As your storage environment grows, operational complexity increases, and knowledge gaps may emerge, especially when key team members transition out.

IBM's global network of field experts and comprehensive knowledge base

Global network of
IBM field experts for
support

Comprehensive
knowledge base for
troubleshooting

Guidance for new
deployments and
migrations

Expert-led assistance
during live incidents



Migrating from Community Ceph to IBM Storage Ceph

Migrating from Community Ceph to IBM Storage Ceph requires more than simply replacing binaries.

The process involves assessing hardware compatibility, reviewing current feature usage, verifying operating system versions, and planning backup and restore strategies to ensure a seamless and risk-managed transition.



Key Steps in IBM's Ceph Migration Methodology

1

Comprehensive
system health
checks and
assessments

2

Detailed feature and
hardware
compatibility
reviews

3

Coordinated OS
and software
upgrade processes

4

Thorough backup
and restore strategy
planning

Business risks of running unsupported storage

The business risks of running unsupported storage are significant.

With Community Ceph, your organization is solely responsible for uptime, compliance, and maintaining data integrity.

If issues arise, all troubleshooting, root cause analysis, and recovery fall entirely on your internal teams, increasing exposure.

Community Ceph

Self

IBM Storage Ceph

IBM



Compliance risks

Running Community Ceph may expose organizations to compliance risks.

Without formal vendor support, you lack official attestations and audit-ready documentation, making regulatory audits more challenging.

Your internal teams must independently gather evidence and ensure ongoing adherence to requirements.

Efficient, Risk-aware Infrastructure with IBM Storage Ceph

Audit-ready logs
for compliance
reporting

Built-in integration
with Active Directory
and Kerberos

Policy-driven
management for
security enforcement

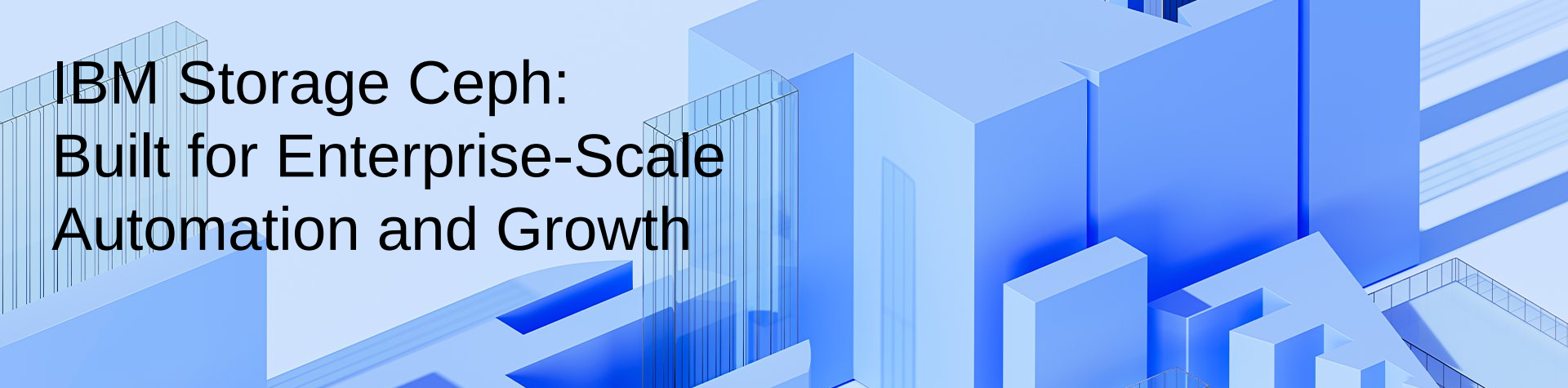
Certified with key
regulatory
frameworks



Scalability challenges

Scalability challenges increase with Community Ceph, as your team must develop and maintain custom scripts for scaling, disaster recovery, and managing multi-region deployments.

Each new business requirement adds engineering overhead and raises the risk of errors during critical operations.



IBM Storage Ceph: Built for Enterprise-Scale Automation and Growth

1

Automated scaling
and lifecycle
management

2

Policy-driven data
stewardship for
governance

3

Seamless
integration with AI
and data
lakehouses

4

Reduces
operational risk as
your needs grow

Maintaining Community Ceph

Maintaining Community Ceph diverts valuable engineering time from business innovation.

The DIY approach consumes skilled resources with maintenance tasks, limiting your team's ability to focus on projects that deliver real strategic value and competitive advantage.

IBM Storage Ceph

IBM Storage Ceph minimizes operational risk, guarantees compliance, offers enterprise-grade support and accelerates value delivery.

Teams can focus on strategic goals instead of infrastructure maintenance.



Enterprise-Grade Reliability



Strategic partnership backed by expert support

IBM Storage Ceph delivers more than just robust technology—it offers a strategic partnership backed by expert support and ongoing innovation.

Designed for the evolving needs of enterprises, it provides stability and confidence, setting a new standard for business-critical storage solutions.

Thank you

